

Lab: CREATE, ALTER, TRUNCATE, DROP Tables

Estimated time needed: 15 minutes

In this lab, you will learn some commonly used DDL (Data Definition Language) statements of SQL. First you will learn the CREATE statement, which is used to create a new table in a database. Next, you will learn the ALTER statement which is used to add, delete, or modify columns in an existing table. Then, you will learn the TRUNCATE statement which is used to remove all rows from an existing table without deleting the table itself. Lastly, you will learn the DROP statement which is used to delete an existing table in a database.

How does the syntax of a CREATE statement look?

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,  
    column3 datatype,  
    ....  
);
```

How does the syntax of an ALTER statement look?

```
ALTER TABLE table_name  
ADD COLUMN column_name data_type column_constraint;  
ALTER TABLE table_name  
DROP COLUMN column_name;  
ALTER TABLE table_name  
ALTER COLUMN column_name SET DATA TYPE data_type;  
ALTER TABLE table_name  
RENAME COLUMN current_column_name TO new_column_name;
```

How does the syntax of a TRUNCATE statement look?

```
TRUNCATE TABLE table_name;
```

How does the syntax of a DROP statement look?

```
DROP TABLE table_name;
```

Software Used in this Lab

In this lab, you will use [IBM Db2 Database](#). Db2 is a Relational Database Management System (RDBMS) from IBM, designed to store, analyze and retrieve the data efficiently.

To complete this lab you will utilize a Db2 database service on IBM Cloud. If you did not already complete this lab task earlier in this module, you will not yet have access to Db2 on IBM Cloud, and you will need to follow this lab first:

- [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console](#)

Database Used in this Lab

The databases used in this lab are internal databases.

Objectives

After completing this lab, you will be able to:

- Create a new table in a database
- Add, delete, or modify columns in an existing table
- Remove all rows from an existing table without deleting the table itself
- Delete an existing table in a database

Instructions

When you approach the exercises in this lab, follow the instructions to run the queries on Db2:

- Go to the [Resource List](#) of IBM Cloud by logging in where you can find the Db2 service instance that you created in a previous lab under **Services** section. Click on the **Db2-xx service**. Next, open the Db2 Console by clicking on **Open Console** button. Click on the 3-bar menu icon in the top left corner and go to the **Run SQL** page. The Run SQL tool enables you to run SQL statements.
 - If needed, follow [Hands-on Lab : Sign up for IBM Cloud, Create Db2 service instance and Get started with the Db2 console](#)

Exercise 1: CREATE

In this exercise, you will use the CREATE statement to create two new tables using Db2.

1. You need to create two tables, **PETSALE** and **PET**. To create the two tables PETSALe and PET, copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. In the **History** section below the editor box, you will be able to see if the query has been executed successfully or not.

```
CREATE TABLE PETSALe (  
  ID INTEGER NOT NULL,  
  PET CHAR(20),  
  SALEPRICE DECIMAL(6,2),  
  PROFIT DECIMAL(6,2),  
  SALEDATe DATE  
);  
  
CREATE TABLE PET (  
  ID INTEGER NOT NULL,  
  ANIMAL VARCHAR(20),  
  QUANTITY INTEGER  
);
```

The screenshot shows the IBM Db2 on Cloud web interface. On the left, a sidebar contains icons for navigation, with the 'SQL' icon highlighted by a red box. The main area is divided into two panes: 'Data objects' on the left and a code editor on the right. The 'Data objects' pane shows a search bar and a list of objects, including 'DMT80331'. The code editor pane has a tab titled '*Untitled ...' and contains SQL code for creating two tables: 'PETALE' and 'PET'. The 'History' panel at the bottom right, also highlighted by a red box, shows a list of executed scripts with columns for 'Script', 'Date', and 'Status'.

Data objects | Saved objects

Find objects

SQL

DMT80331

***Untitled ...**

```

1 CREATE TABLE PETALE (
2   ID INTEGER NOT NULL,
3   PET CHAR(20),
4   SALEPRICE DECIMAL(6,2),
5   PROFIT DECIMAL(6,2),
6   SALEDATE DATE
7 );
8
9 CREATE TABLE PET (
10  ID INTEGER NOT NULL,
11  ANIMAL VARCHAR(20),
12  QUANTITY INTEGER
13 );

```

History

Find history

Script	Date	Status
Untitled - 1	Apr 21, 2023 4:03:57 PM	✓ 2
CREATE TABLE PETALE (ID INTEGER NOT NULL, PET CHAR(20), SALEPRIC...		✓
CREATE TABLE PET (ID INTEGER NOT NULL, ANIMAL VARCHAR(20), QUANTI...		✓

2. Now insert some records into the two newly created tables and show all the records of the two tables. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```

INSERT INTO PETALE VALUES
(1, 'Cat', 450.09, 100.47, '2018-05-29'),
(2, 'Dog', 666.66, 150.76, '2018-06-01'),
(3, 'Parrot', 50.00, 8.9, '2018-06-04'),
(4, 'Hamster', 60.60, 12, '2018-06-11'),
(5, 'Goldfish', 48.48, 3.5, '2018-06-14');

```

```

INSERT INTO PET VALUES
(1, 'Cat', 3),
(2, 'Dog', 4),
(3, 'Hamster', 2);

```

```

SELECT * FROM PETALE;
SELECT * FROM PET;

```

The screenshot shows the IBM Db2 on Cloud interface. On the left, there's a sidebar with icons for Data objects, Saved objects, and SQL. The main area is divided into two panes. The top pane, titled 'Data objects', contains a search bar 'Find objects' and a list of objects including 'DMT80331'. The bottom pane, titled 'SQL', contains a code editor with the following SQL script:

```

1 INSERT INTO PETSale VALUES
2   (1, 'Cat', 450.09, 100.47, '2018-05-29'),
3   (2, 'Dog', 666.66, 150.76, '2018-06-01'),
4   (3, 'Parrot', 50.00, 8.9, '2018-06-04'),
5   (4, 'Hamster', 60.60, 12, '2018-06-11'),
6   (5, 'Goldfish', 48.48, 3.5, '2018-06-14');
7
8 INSERT INTO PET VALUES
9   (1, 'Cat', 3),
10  (2, 'Dog', 4),
11  (3, 'Parrot', 2),
12  (4, 'Hamster', 1),
13  (5, 'Goldfish', 1);

```

Below the code editor, there's a 'History' section with a search bar 'Find history' and a table of executed scripts:

Script	Date	Status
Untitled - 1	Apr 21, 2023 4:08:05 PM	✓ 4
INSERT INTO PETSale VALUES (1, 'Cat', 450.09, 100.47, '2018-05-29'), (...)		✓
INSERT INTO PET VALUES (1, 'Cat', 3), (2, 'Dog', 4), (3, 'Hamster', 2)		✓
SELECT * FROM PETSale		✓
SELECT * FROM PET		✓

You can click on the query in the History section to check its result:

The screenshot shows the IBM Db2 on Cloud interface. On the left, there's a sidebar with icons for Data objects, Saved objects, and SQL. The main area is divided into two panes. The top pane, titled 'Data objects', contains a search bar 'Find objects' and a list of objects including 'DMT80331'. The bottom pane, titled 'SQL', contains the same SQL script as in the previous screenshot. Below the code editor, there's a 'Results' section with a search bar 'Filter table' and a table of results:

ID	PET	SALEPRICE	PROFIT
1	Cat	450.09	100.47
2	Dog	666.66	150.76
3	Parrot	50.00	8.90
4	Hamster	60.60	12.00
5	Goldfish	48.48	3.50

Exercise 2: ALTER

In this exercise, you will use the ALTER statement to add, delete, or modify columns in two of the existing tables created in exercise 1.

Task A: ALTER using ADD COLUMN

1. Add a new **QUANTITY** column to the **PETSale** table and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```

ALTER TABLE PETSale
ADD COLUMN QUANTITY INTEGER;
SELECT * FROM PETSale;

```

The screenshot shows the IBM Db2 on Cloud console interface. On the left, there is a sidebar with icons for navigation. The main area is divided into two tabs: "Data objects" and "Saved objects". Under "Data objects", there is a search bar labeled "Find objects" and a list of objects, including "DMT80331". The right side of the console features a SQL editor with a tab labeled "*Untitled ...". The editor contains the following SQL code:

```
1 ALTER TABLE PETSale
2 ADD COLUMN QUANTITY INTEGER;
3
4 SELECT * FROM PETSale;
```

Below the editor, there is a "Results" section showing "Result set 1". It includes a "Filter table" search bar and a table with the following data:

ID	PET	SALEPRICE	PROFIT	SALED
1	Cat	450.09	100.47	2018-01-01
2	Dog	666.66	150.76	2018-01-01
3	Parrot	50.00	8.90	2018-01-01
4	Hamster	60.60	12.00	2018-01-01
5	Goldfish	48.48	3.50	2018-01-01

2. Now update the newly added **QUANTITY** column of the **PETSale** table with some values and show all the records of the table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. After the query has executed successfully, click on it to check the result set.

```
UPDATE PETSale SET QUANTITY = 9 WHERE ID = 1;
UPDATE PETSale SET QUANTITY = 3 WHERE ID = 2;
UPDATE PETSale SET QUANTITY = 2 WHERE ID = 3;
UPDATE PETSale SET QUANTITY = 6 WHERE ID = 4;
UPDATE PETSale SET QUANTITY = 24 WHERE ID = 5;
SELECT * FROM PETSale;
```

IBM Db2 on Cloud

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SQL

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Data objects

Saved objects

🔍 Find objects

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🗃️ DMT80331

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🟢 Syntax assi

1 UPDATE PETSALE SET QUANTITY = 9 WHERE ID = 1;

2 UPDATE PETSALE SET QUANTITY = 3 WHERE ID = 2;

3 UPDATE PETSALE SET QUANTITY = 2 WHERE ID = 3;

4 UPDATE PETSALE SET QUANTITY = 6 WHERE ID = 4;

5 UPDATE PETSALE SET QUANTITY = 24 WHERE ID = 5;

6

7 SELECT * FROM PETSALE;

History

Results

Result set 1

Details

🔍 Filter table

ID	PET	SALEPRICE	PROFIT	SALED
1	Cat	450.09	100.47	2018-(
2	Dog	666.66	150.76	2018-(
3	Parrot	50.00	8.90	2018-(
4	Hamster	60.60	12.00	2018-(
5	Goldfish	48.48	3.50	2018-(

Task B: ALTER using DROP COLUMN

1. Delete the **PROFIT** column from the **PETSALE** table and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

ALTER TABLE PETSALE
DROP COLUMN PROFIT;
SELECT * FROM PETSALE;

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Task C: ALTER using ALTER COLUMN

1. Change the data type to **VARCHAR(20)** type of the column **PET** of the table **PETSALE** and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```
ALTER TABLE PETALE
ALTER COLUMN PET SET DATA TYPE VARCHAR(20);
SELECT * FROM PETALE;
```

The screenshot shows the IBM Db2 on Cloud interface. On the left, the 'Data objects' tab is active, displaying a search bar and a list of objects including 'DMT80331'. The main panel shows a SQL editor with the following code:

```
1 ALTER TABLE PETSale
2 ALTER COLUMN PET SET DATA TYPE VARCHAR(20);
3
4 SELECT * FROM PETSale;
```

The 'Results' tab is active, showing 'Result set 1' with a table of data. The table has four columns: ID, PET, SALEPRICE, and SALEDATE. The data is as follows:

ID	PET	SALEPRICE	SALEDATE
1	Cat	450.09	2018-05-29
2	Dog	666.66	2018-06-01
3	Parrot	50.00	2018-06-04
4	Hamster	60.60	2018-06-11
5	Goldfish	48.48	2018-06-14

2. Now verify if the data type of the column **PET** of the table **PETSale** changed to **VARCHAR(20)** type or not. Click on the Data Section in the left menu bar.

The screenshot shows the IBM Db2 on Cloud interface. The left menu bar has the 'Data' section highlighted with a red box. The main panel shows the 'Data objects' tab with a search bar and a list of objects including 'DMT80331'.

Then click on Tables:

The screenshot shows the IBM Db2 on Cloud interface. The top navigation bar has the 'Tables' section highlighted with a red box. The main panel shows the 'Load Data' section with a 'Source' and 'Target' field. The 'Source' field is labeled 'You are loading the file' and the 'Target' field is labeled 'File selection'. Below the 'Source' field, there is a section for 'My Computer' with the text: 'A single delimited text file (CSV) without header row.'

Find your schema and choose the table **PETSALE**

IBM Db2 on Cloud

Load Data Load History **Tables** Views Indexes Aliases MQTs Sequences Application objects

Find schemas or tables

Schemas

☒ Name	Definer type	Tables ▲
☒ DMT80331	User	3

Total: 1, selected: 1

Tables

☐ Name ▼	Schema
☐ BILLING_TEST	DMT80331
☐ PET	DMT80331
☐ PETSALE	DMT80331

Total: 3, selected: 0

You will see that the datatype of the column **PET** has changed to **VARCHAR(20)**

IBM Db2 on Cloud

Load Data Load History **Tables** Views Indexes Aliases MQTs Sequences Application objects

Find schemas or tables

Tables

New table +

☐ Name ▼	Schema	Properties
☐ BILLING_TEST	DMT80331	...
☐ PET	DMT80331	...
☐ PETSALE	DMT80331	...

Total: 3, selected: 0

Table definition

PETSALE

Name	Data type	Nullabl
ID	INTEGER	N
PET	VARCHAR	Y
SALEPRICE	DECIMAL	Y
SALEDATE	DATE	Y
QUANTITY	INTEGER	Y

View data

Task D: ALTER using RENAME COLUMN

1. In the **PETSALE** table, rename the column **PET** to **ANIMAL** and show the altered table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**.

```
ALTER TABLE PETSALE
RENAME COLUMN PET TO ANIMAL;
SELECT * FROM PETSALE;
```

The screenshot shows the IBM Db2 on Cloud console interface. On the left, a sidebar contains icons for navigation. The main area is divided into two panes. The left pane, titled 'Data objects', shows a search bar and a list of objects, including 'DMT80331'. The right pane, titled '*Untitled ...', displays a SQL query editor with the following code:

```
1 ALTER TABLE PETSale
2 RENAME COLUMN PET TO ANIMAL;
3
4 SELECT * FROM PETSale;
```

Below the query editor, there are tabs for 'History' and 'Results'. The 'Results' tab is active, showing 'Result set 1' with a 'Details' link. Below this is a table with the following data:

ID	ANIMAL	SALEPRICE	SALEDATE
1	Cat	450.09	2018-05-
2	Dog	666.66	2018-06-
3	Parrot	50.00	2018-06-
4	Hamster	60.60	2018-06-
5	Goldfish	48.48	2018-06-

In this exercise, you will use the TRUNCATE statement to remove all rows from an existing table created in exercise 1 without deleting the table itself.

1. Remove all rows from the **PET** table and show the empty table. Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. You will see **no data in the Result section**.

```
TRUNCATE TABLE PET IMMEDIATE;
SELECT * FROM PET;
```

The screenshot shows the IBM Db2 on Cloud console. On the left, the 'Data objects' tab is active, displaying a search bar and a list of objects including 'DMT80331'. The main area is the SQL editor, titled '*Untitled ...', containing the following SQL code:

```
1 TRUNCATE TABLE PET IMMEDIATE;
2
3 SELECT * FROM PET;
```

Below the editor, the 'Results' pane is visible, showing 'Result set 1' with a 'Filter table' search bar. The table structure is displayed with columns: ID, ANIMAL, and QUANTITY. Below the table structure, a message states: 'You don't have any data currently'.

In this exercise, you will use the DROP statement to delete an existing table created in exercise 1.

1. Delete the **PET** table and verify if the table still exists or not (SELECT statement won't work if a table doesn't exist). Copy the code below and paste it to the textbox of the **Run SQL** page. Click **Run all**. You will see that the **select statement fails**.

```
DROP TABLE PET;
SELECT * FROM PET;
```

The screenshot shows the IBM Db2 on Cloud console. The SQL editor now contains the following SQL code:

```
1 DROP TABLE PET;
2
3 SELECT * FROM PET;
```

Below the editor, the 'History' pane is visible, showing a table of executed scripts. The table has columns: Script, Date, and Status. The first two rows show successful execution of 'Untitled - 1' and 'DROP TABLE PET'. The third row, 'SELECT * FROM PET', is highlighted with a red border and shows a failure status (red circle with a slash).

Script	Date	Status
Untitled - 1	Apr 21, 2023 4:28:00 PM	✓ 1
DROP TABLE PET		✓
SELECT * FROM PET		✗

Congratulations! You have completed this Lab. You are ready for the next topic.

Author(s)

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